Water Purification
(Different Types of Water Identifier – ML & AI)

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Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

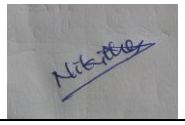
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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT (Maximum 300 words)

Water quality monitoring plays a critical role in ensuring safe drinking water and protecting environmental ecosystems. Conventional water testing methods often require laboratory analysis, specialized instruments, and trained personnel, making them time-consuming and expensive. As a result, continuous and real-time water monitoring becomes difficult in many situations. This research proposes an Artificial Intelligence (AI) and Machine Learning (ML) based water identification system capable of automatically determining different types of water using sensor-generated data.

The system will utilize multiple water quality sensors to measure parameters such as pH, Total Dissolved Solids (TDS), turbidity, conductivity, and temperature. These parameters provide valuable information about the chemical and physical characteristics of water. The collected data will then be processed using machine learning algorithms that can identify patterns and relationships within the dataset. Based on these patterns, the system will classify water into categories such as distilled water, tap water, alkaline water, mineral water, and potentially contaminated water.

By integrating sensor technologies with machine learning models, the system aims to provide rapid, reliable, and automated water identification. This approach can significantly reduce dependence on manual testing methods and laboratory analysis while improving efficiency in water quality monitoring. The proposed system has potential applications in households, industries, environmental monitoring, and public health management.

Keywords: Alkaline Water, Water Purification, IoT (Internet of Things), Artificial Intelligence (AI), Machine Learning, Mineral Balancing, pH Monitoring, Total Dissolved Solids (TDS), Automated Dosing, Smart Water System

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LIST OF ABBREVIATIONS

Abbrevations	Full Form	Explanation
AI	Artificial Intelligence	The branch of computer science that enables machines to mimic human intelligence, including learning, reasoning, problem-solving, and decision-making. AI is used in applications like natural language processing, robotics, and smart systems.
ML	Machine Learning	A subset of AI that focuses on building systems that can learn from data and improve their performance over time without explicit programming. ML is widely used for predictions, classification, and pattern recognition.
IoT	Internet of Things	A network of interconnected physical devices that collect and exchange data over the internet. IoT devices include sensors, smart appliances, and wearable tech, enabling automation and remote monitoring.
TDS	Total Dissolved Solids	A measure of the combined content of all inorganic and organic substances (such as minerals, salts, and metals) dissolved in water. High TDS levels can affect water quality, taste, and safety.
pH	Potential of Hydrogen	A scale used to measure the acidity or alkalinity of a solution, ranging from 0 (very acidic) to 14 (very alkaline), with 7 being neutral. pH is crucial in water quality and chemical processes.
SVM	Support Vector Machine	A supervised machine learning algorithm used for classification and regression tasks. SVM works by finding the optimal hyperplane that separates data points of different classes.
ANN	Artificial Neural Network	A computational model inspired by the human brain, consisting of interconnected nodes (neurons). ANNs are used in deep learning for tasks like image recognition, speech processing, and predictive modeling.
KNN	K-Nearest Neighbors	A simple, non-parametric machine learning algorithm used for classification and regression. KNN predicts the output of a data point based on the majority label or average of its nearest neighbors in the feature space.

Table 1 List of Abbreviations and Their Explanations

- The above abbreviations represent key concepts and technologies used in modern intelligent systems, particularly in areas like water quality monitoring, data analysis, and smart IoT applications. Understanding these terms provides a foundational knowledge for implementing advanced solutions, integrating AI and ML techniques, and effectively interpreting water quality parameters such as TDS and pH. These concepts collectively support the development of efficient, automated, and intelligent water purification systems.

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1. INTRODUCTION

1.1 Background and Context

Access to clean and safe drinking water is one of the most fundamental requirements for human health and sustainable development. However, water sources around the world vary greatly in quality due to natural factors and human activities such as industrial discharge, agricultural runoff, and urban pollution. As a result, it is essential to continuously monitor and evaluate water quality to ensure its safety for consumption and other uses**[6]**.

Traditional water quality testing methods typically involve collecting water samples and analyzing them in laboratories using chemical analysis techniques**[1]**. While these methods are accurate, they require specialized equipment, trained personnel, and significant time. In many cases, this makes frequent monitoring difficult and expensive.

With the advancement of modern technologies such as Artificial Intelligence, Machine Learning, and Internet of Things (IoT), new opportunities have emerged for developing automated water monitoring systems. Machine learning algorithms can analyze large datasets and identify patterns that may not be easily detected through traditional analytical methods. **By integrating sensor networks with machine learning models**, water quality parameters can be measured in real time and used to automatically classify different types of water**[2]**.

The proposed research focuses on **developing an intelligent system capable of identifying water types based on sensor data**. The system will combine hardware components such as sensors and microcontrollers with software components such as **machine learning models and data processing algorithms.** This integration will enable real-time monitoring, automated classification, and efficient decision-making related to water quality.



Figure 1 Traditional Sand–Charcoal Filtration Method for Water Purification

1.2 Literature Review

The convergence of low-cost sensors, IoT, and machine learning has opened new frontiers in environmental monitoring.

- **Traditional and IoT-Based Monitoring:** Early work focused on manual sampling and laboratory analysis, which remains the benchmark for accuracy but is impractical for continuous monitoring **[1]**. The advent of IoT enabled the creation of real-time monitoring stations. **Kumar & Singh (2021)** demonstrated an IoT-based system using an Arduino and multiple sensors to monitor water quality parameters and upload data to the cloud**[2]**. These systems provide continuous data but are typically designed for threshold-based alerts (e.g., "pH too high") rather than sophisticated classification.Similar IoT monitoring frameworks have also been proposed for environmental monitoring using wireless sensor networks **[8]**, **[10]**.

- **Machine Learning for Water Quality Analysis:** Researchers have increasingly applied ML to water quality data. Uddin et al. (2020) conducted a comprehensive study comparing various ML algorithms including Random Forest, Decision Trees, and Support Vector Machines (SVM) for predicting water quality indices**[4]**. Their findings showed that ensemble methods like Random Forest often outperform simpler models in accuracy. Other studies have used Artificial Neural Networks (ANNs) to model complex, non-linear relationships in water quality parameters, such as predicting dissolved oxygen levels **[3]**. Additional studies have also explored machine learning approaches for predicting water quality parameters and environmental monitoring tasks **[9]**.

- **Focus on Classification:** The specific task of classifying water into discrete types (e.g., distilled, tap, alkaline) is less explored. Some research exists on using electronic tongues (e-tongues) arrays of chemical sensors combined with pattern recognition methods for classifying beverages and water**[5]**. These systems are often complex and designed for laboratory use. The proposed research aims to simplify and focus on this concept by using a standard set of physicochemical sensors and a purpose-built ML model to create a practical, low-cost water "type" classifier for real-world applications.

  Water quality assessment studies also frequently rely on the **Water Quality Index (WQI)** to evaluate overall water safety and environmental status **[7]**.

1.3 Research Gap

The academic and commercial landscape of water quality monitoring has undoubtedly advanced, transitioning from purely manual methods to sophisticated cyber-physical systems. Recent literature is rich with examples of Internet of Things (IoT) sensor networks enabling real-time data acquisition, and Machine Learning (ML) models demonstrating high accuracy in predicting water quality indices. However, a critical examination reveals that these

advancements, while significant, have largely converged on a narrow objective: **anomaly detection**. This focus on identifying deviations from a norm, while useful for issuing warnings, falls short of providing the **generative, categorical intelligence** required for automated environmental management and intervention. The true gap lies in the transition from a system that can *shout "something is wrong"* to one that can *reason "this is what it is, and therefore this is what must be done."*

This research posits that a fundamental, yet unaddressed, gap exists in the **classification-centric paradigm** of water monitoring. This gap can be deconstructed into three primary dimensions:

1. The Conceptual Gap: From Binary Alerts to Categorical Intelligence.

The prevailing paradigm in smart water monitoring is threshold-based anomaly detection. A system is programmed with safety limits (e.g., pH 6.5–8.5), and an alert is triggered when a reading falls outside this range. While this serves a basic public health warning function, it is a binary, reactive, and low-information output. It tells a user that water is *not safe*, but it fails to tell them *what it is*.

This represents a critical conceptual gap. In numerous applied contexts, the question is not "is this water safe?", but rather "what category of water is this?". For instance:

- In an **automated industrial or agricultural setting**, a system needs to distinguish between freshwater, brackish water, and saline water to adjust desalination or mineral dosing processes accordingly.

- In a **water treatment plant**, identifying raw source water as "highly turbid with alkaline pH" versus "moderately turbid with neutral pH" triggers entirely different pre-treatment and coagulation protocols.

- In **environmental forensics**, classifying a sample as "agricultural runoff" versus "industrial effluent" based on its combined chemical signature is far more valuable than a simple "contamination detected" alert.

Literature lacks a robust, practical focus on systems designed for this kind of categorical classification. The distinction is not semantic; it is functional. An anomaly alert requires human intervention to diagnose the problem, whereas a classification output can be the direct input for an automated, context-aware response.

2. The Systemic Gap: Siloed Hardware and Disembodied Intelligence.

A second dimension of the gap lies in the lack of genuine, end-to-end integration. Many studies demonstrate powerful ML models for water classification, but they are typically trained and validated on historical or carefully curated datasets, disconnected from the physical world. Conversely, numerous IoT hardware projects excel at real-time data acquisition but treat data analysis as a separate, post-processing step, often relying on simple threshold logic.

There is a distinct absence of a **seamless, closed-loop pipeline** that unites these two domains. The gap is the missing interface where raw, real-time, and potentially noisy sensor data is fed directly into a trained ML model running on an edge device or a unified backend. This systemic fragmentation means that “intelligence" remains disembodied as an academic exercise rather than an operational tool. The challenge is not just building a smart model but embedding it within a live data stream to create a system that can classify water *in situ* and in real-time.

3. The Actionability Gap: The Missing "Smart Water Brain".

The final and most critical gap is the absence of a unified platform that closes the loop from sensing to action. Current solutions are siloed: a telemetry system for data collection, a separate software environment for analysis, and a manual process for decision-making and intervention. This fragmented architecture creates latency and inefficiency, preventing true proactive management.

What is missing is a **unified platform "smart water brain” that integrates three core functions into a cohesive whole**:

1. **Sensory Input:** Direct, real-time ingestion of data from a multi-parameter sensor array.
2. **Cognitive Processing:** An embedded, trained ML model that instantly processes this data stream to perform categorical classification (e.g., "Class: Alkaline, Turbidity: High, Salinity: Low").
3. **Executive Output:** The ability to translate that classification into an automated, pre-defined action. This action could be a digital output (e.g., sending a detailed alert to a user dashboard with the water class and recommended action) or a physical one (e.g., triggering a relay to adjust a dosing pump, open a specific valve, or command an autonomous vehicle to loiter for further sampling).

This research directly confronts this tripartite gap. It proposes the development of an **end-to-end intelligent system** that moves beyond simple anomaly detection to achieve true categorical classification. By architecting a unified platform that integrates real-time multi-parameter sensing, a curated and relevant dataset, a robust ML classifier, and an automated action-triggering interface, this project will deliver a practical, powerful, and generative tool, a "smart water brain" capable of not just monitoring water but understanding it and acting upon that understanding.

2. OBJECTIVE

2.1 Main Objective

To design, develop, and validate an **intelligent water identification system** that leverages machine learning models to automatically and accurately classify water types (e.g., distilled, tap, alkaline, contaminated) based on real-time data from a multi-sensor array.

2.2 Specific Objectives

1. **Hardware Design and Data Acquisition System:** To design and assemble a reliable data acquisition system by integrating an array of water quality sensors (pH, TDS, Turbidity, Conductivity, Temperature) with a microcontroller (e.g., ESP32 or Arduino) and develop the firmware to collect and log synchronized sensor readings.

2. **Dataset Creation and Curation:** To collect a robust, high-quality dataset by sampling multiple distinct water types (distilled, tap, mineral, alkaline, synthetic contaminated) under controlled conditions. This will involve creating a detailed data collection protocol, ensuring many samples per class, and accurately labeling the data for supervised learning.

3. **Model Development and Training:** To preprocess the collected dataset (handling missing values, normalization) and train, validate, and compare the performance of several supervised machine learning classification algorithms (e.g., Logistic Regression, K-Nearest Neighbors, Support Vector Machine, Random Forest, Neural Network) to identify the most accurate and efficient model for the task.

4. **System Integration and Real-Time Inference:** To integrate the trained ML model into the data acquisition pipeline, creating a system that can take live sensor readings, pass them to the model, and return a water type classification in near real-time.

5. **User Interface and Visualization:** To develop a simple, intuitive user interface (e.g., a desktop application, a basic web dashboard, or an LCD display) that displays the real-time sensor values, the final water type classification, and a timestamp of the analysis.

6. **Model Evaluation and System Validation:** To rigorously evaluate the final integrated system's performance in a lab setting by testing it on new, unseen water samples and comparing its classification accuracy against known labels, thereby validating its reliability and effectiveness.

3. METHODOLOGY

The development will follow a structured, iterative process combining hardware and software engineering.

- **Phase 1: Hardware Setup and Firmware Development**
    - **Component Selection:** An ESP32 microcontroller is ideal due to its built-in Wi-Fi/Bluetooth, analog inputs, and processing power. Sensors will include a gravity analog pH sensor, a TDS meter, a turbidity sensor, a DS18B20 waterproof temperature sensor, and a conductivity sensor.
    - **Circuit Assembly:** Sensors will be connected to the ESP32 via a breadboard or custom PCB. Voltage dividers or signal conditioning circuits will be used if necessary to ensure sensor readings are within the ESP32's 0-3.3V range.
    - **Firmware (Arduino IDE):** Code will be written to initialize sensors, read analog/digital values at a set frequency (e.g., once per second), convert raw readings to meaningful units (pH, ppm, NTU, µS/cm, °C), and print this data to the serial monitor for logging. A data packet structure will be defined for consistent formatting.
- **Phase 2: Data Collection and Preprocessing**
    - **Sampling Protocol:** For each water type (Distilled, Tap, Alkaline, Mineral, Contaminated), multiple samples will be collected. For each sample, the sensor array will take readings for a sustained period (e.g., 2-3 minutes) to capture variability. All readings will be logged to a CSV file with a label column.
    - **Dataset Preparation:** The raw CSV will be loaded into a Python environment (Pandas). Data will be cleaned (remove outliers), checked for null values, and normalized/standardized (e.g., using StandardScaler from scikit-learn) to ensure all features contribute equally to model training.
- **Phase 3: Machine Learning Model Development**
    - **Data Splitting:** The labeled dataset will be split into training (70-80%), validation (10-15%), and testing (10-15%) sets.
    - **Model Selection & Training:** Several classification algorithms from the scikit-learn library will be trained on the training data:
        - **Baseline Models:** Logistic Regression, K-Nearest Neighbors (KNN).
        - **Ensemble Models:** Random Forest, Gradient Boosting.
        - **Kernel-Based Model:** Support Vector Machine (SVM).

        - **Neural Network:** A simple Multi-Layer Perceptron (MLP) using TensorFlow/Keras.
    - **Hyperparameter Tuning:** GridSearchCV or RandomizedSearchCV will be used to find the optimal hyperparameters for each model based on performance on the validation set.
    - **Model Evaluation & Selection:** Models will be evaluated on the unseen test set using metrics like accuracy, precision, recall, and F1-score. Confusion matrices will be analyzed to see which water types are most misclassified. The best-performing model (balancing accuracy and speed) will be selected.

- **Phase 4: System Integration and Deployment**
    - **Model Export:** The best-performing trained model will be serialized (e.g., using Python's pickle or joblib) so it can be loaded without retraining.
    - **Inference Pipeline:** A Python script will be created to:
        1. Read a new data packet from the ESP32 (via serial or Wi-Fi).
        2. Apply the same preprocessing/normalization used during training.
        3. Load the serialized ML model.
        4. Pass the preprocessed sensor values to the model and obtain a predicted water type.
        5. Output the result.

- **Phase 5: User Interface and Validation**
    - **UI Development:** A simple real-time dashboard will be created using libraries like Tkinter (for desktop) or Streamlit (for web). It will display live sensor gauges and the current water type prediction.
    - **Final System Validation:** The complete system will be tested on a final set of blind water samples (not used in training or testing) to confirm its real-world accuracy and response time.

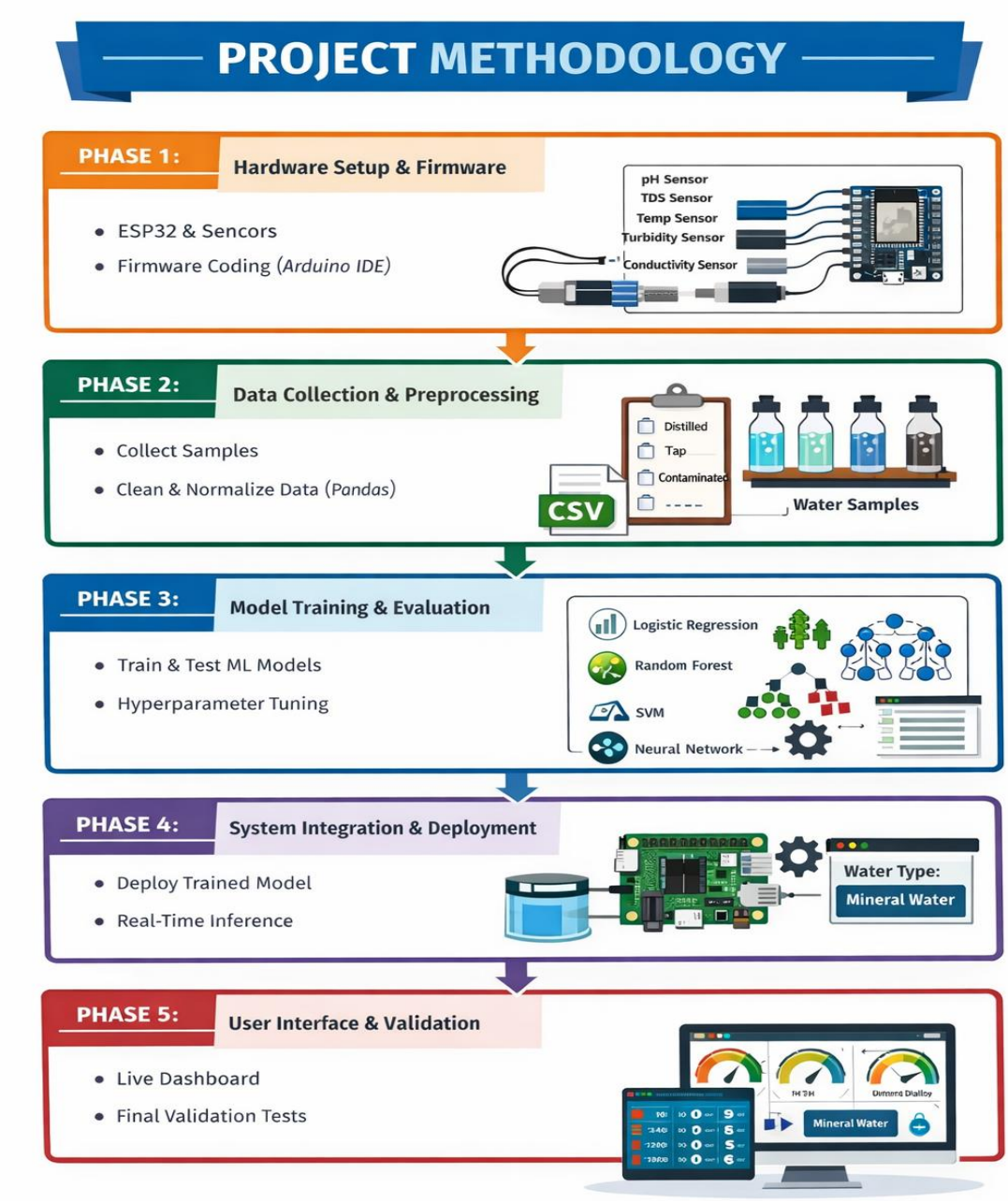


Figure 2 Overview of the Proposed AI-Based Water Identification System Methodology

4. HIGH-LEVEL SYSTEM ARCHITECTURE

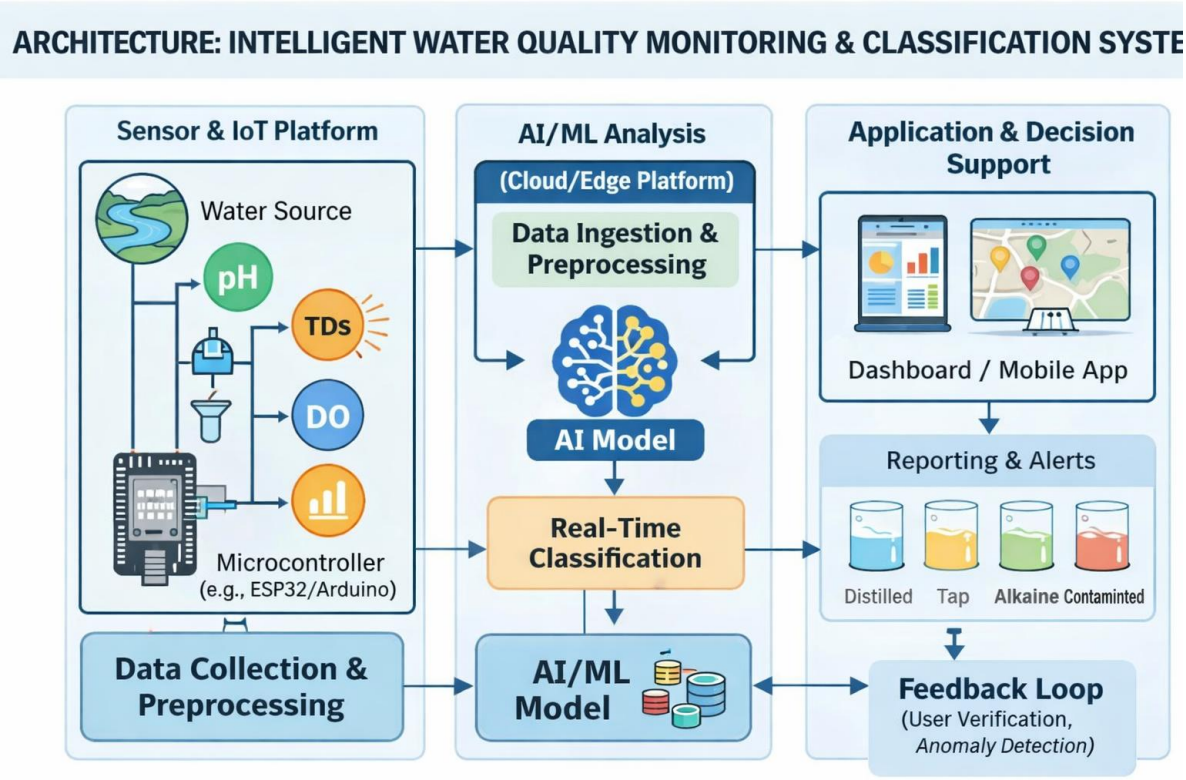


Figure 3 High-level architecture of the intelligent water quality monitoring and classification system, showing sensor data collection, AI/ML analysis, and user-facing dashboards for real-time insights.

- The system integrates **IoT sensors, microcontrollers, and AI models** to classify water quality in real time. Sensors such as pH, turbidity, TDS, and temperature capture essential parameters from the water source. A microcontroller (e.g., ESP32/Arduino) collects and preprocesses this data before transmitting it to an **AI/ML model**. The model analyzes the inputs and classifies water into categories like **Distilled, Tap, Alkaline, or Contaminated**. Results are displayed on a **dashboard or mobile app**, providing users with actionable insights and alerts. A feedback loop ensures continuous improvement by incorporating user verification and anomaly detection

5. USER REQUIREMENTS

The proposed system should satisfy several functional and non-functional requirements to ensure effective operation.

Functional Requirements

- **Data Acquisition -** The system shall continuously measure and digitize the following water parameters: pH (0-14), TDS (0-1000+ ppm), Turbidity (0-1000+ NTU), Electrical Conductivity (0-2000 µS/cm), and Temperature (0-50°C).
- **Data Processing -** The system shall preprocess raw sensor readings (e.g., apply calibration formulas, filter noise) to produce reliable quantitative values.
- **Classification -** The system shall, upon user request or automatically at defined intervals, use its ML model to classify the water sample into one of the predefined categories (e.g., Distilled, Tap, Alkaline, Contaminated).
- **User Interface -** The system shall provide a user interface that displays current sensor readings, the latest water type classification, and a timestamp of the analysis.
- **Data Logging -** The system shall log all sensor readings and classification results with timestamps to a local file (CSV) for historical tracking.

Non-Functional Requirements

- **Accuracy -** The final ML model's classification accuracy on unseen data must be at least **90%** across all target water types.
- **Latency -** The time from receiving a new sensor data packet to displaying the classification result should be less than **2 seconds**.
- **Reliability):** The system should be robust to minor sensor drift or noise, maintaining reasonable classification accuracy over time without immediate recalibration.
- **NFR-04 (Usability):** The user interface should be simple and intuitive, requiring no technical expertise in ML or sensors to interpret the results ("Traffic light" indicators or clear labels are preferable).
- **NFR-05 (Cost-Effectiveness):** The total bill of materials for the hardware prototype should be kept under **LKR 25,000** to ensure potential affordability for households or small communities.
- **NFR-06 (Modularity):** The software and hardware design should be modular, allowing for easy replacement of sensors or updates to the ML model.

6. COMMERCIALIZATION PLAN

The AI-based water identifier is not just a research project but a product with significant market potential, positioned at the intersection of health, technology, and environmental awareness.

- **Target Market Segments:**
    1. **Consumer Electronics (Smart Homes):** The most direct application is as a standalone "smart water tester" for health-conscious consumers. It could verify the quality of tap water, confirm the claims of expensive bottled alkaline or mineral water, and test water from household filters to signal when filters need replacement.
    2. **Embedded OEM Component:** The core technology (sensors + ML model) can be licensed or integrated as a module ("smart water brain") into existing products.
        - **Water Purifier Companies:** Integrating this module would allow purifiers to become "smart," automatically detecting the incoming water type and adjusting the purification process (e.g., deciding whether to use RO or UV, or how many minerals to add back).
        - **Appliance Manufacturers:** Integration into smart refrigerators with water dispensers to provide real-time quality feedback to the user.
    3. **Industrial and Commercial Monitoring:** Industries that require specific water grades for their processes (e.g., pharmaceutical companies requiring distilled water, breweries, beverage companies) can use the system for instant, inline quality assurance.
    4. **Environmental and Field Agencies:** A portable, battery-powered version could be used by entities like the CEA or NWSDB for rapid field assessments and initial screening of water sources before more detailed lab analysis.
- **Business Model:**
    - **Product Sales:** Direct-to-consumer sales of the "Smart Water Tester" kit through online platforms and retail stores.
    - **B2B Licensing:** Licensing the technology (software + hardware design) to purification and appliance manufacturers for integration into their product lines.
    - **Data and Analytics Service:** Offering a cloud-connected version where users can track their water quality data over time, with premium analytics and insights provided via a subscription.

- **Competitive Advantage:** The key differentiator is the **AI-powered classification** itself. While simple TDS pens exist, they provide only one data point. This system fuses multiple parameters and uses machine learning to provide a holistic, intelligent verdict on the *type* of water, offering a level of insight currently unavailable in a low-cost, consumer-friendly format.

7. BUDGET AND JUSTIFICATION

Component	Quantity	Unit Price (LKR)	Total Cost
ESP32 Microcontroller	1	2,500	2,500
pH Sensor Module	1	1,500	1,500
TDS Sensor	1	1,000	1,000
Turbidity Sensor	1	500	500
Breadboard & Wires	1	1,000	1,000
Power Supply	1	5000	5000
Miscellaneous Components	-	2,000	2,000

Table 1 Estimated project budget allocation across major tasks and phases, showing resource distribution and financial planning.

- Budget Justification - The proposed system aims to develop a cost-effective water identification device. All components were selected based on affordability, availability, and compatibility with microcontroller platforms. The ESP32 microcontroller provides sufficient computational power while maintaining low energy consumption. Sensor modules were chosen to measure key water quality parameters that significantly influence water classification. The overall budget remains within the project's affordability constraint of **13,500 LKR**, making the prototype feasible for household and community deployment.

8. WORK BREAKDOWN STRUCTURE (WBS)

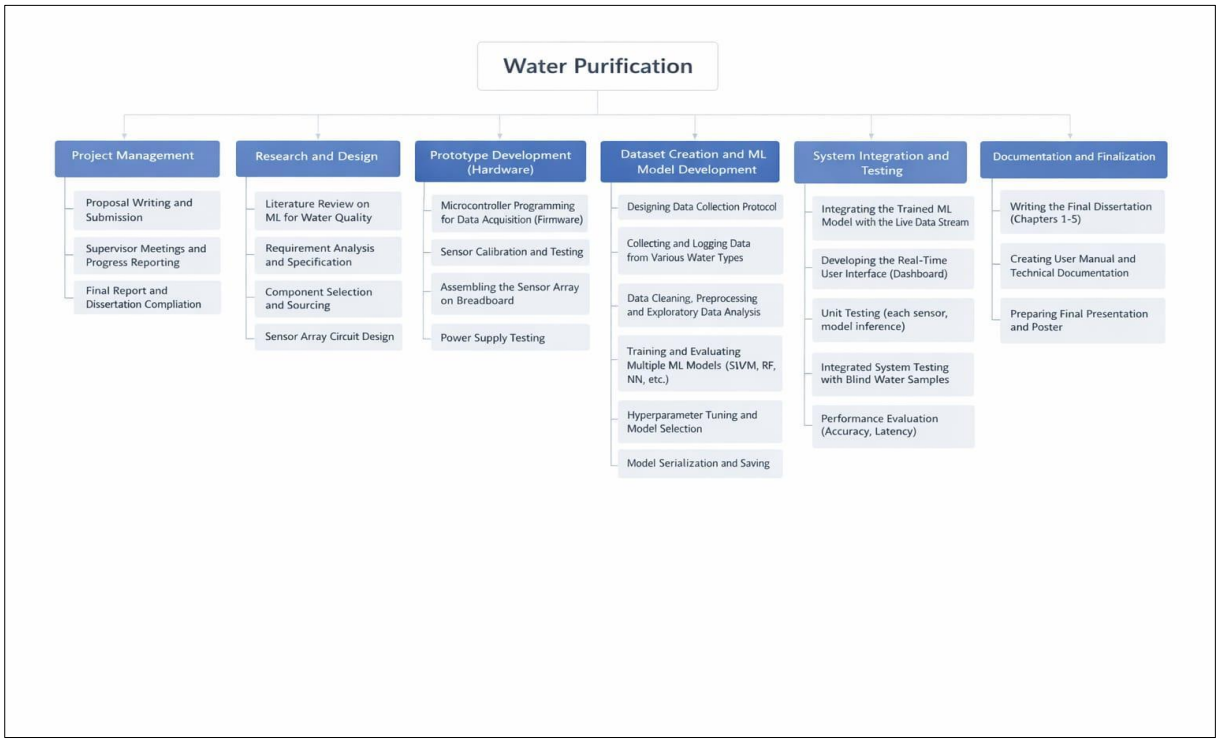


Figure 4 Work Breakdown Structure chart illustrating the hierarchical decomposition of project tasks into manageable phases, ensuring clarity in scope, responsibilities, and deliverables

- The Work Breakdown Structure (WBS) is a hierarchical framework that breaks the project into smaller, manageable components. It ensures clarity in scope, responsibilities, and deliverables by organizing tasks into phases

9. GANTT CHART

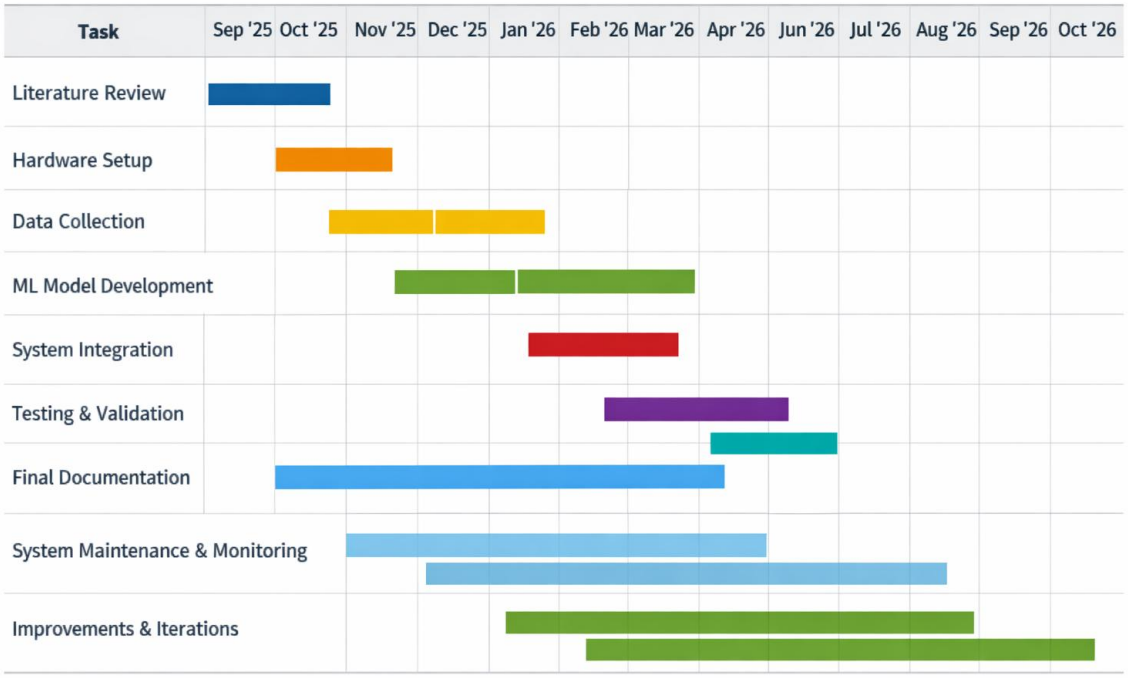


Figure 5 Gantt chart illustrating the project timeline from September 2025 to October 2026, showing task scheduling, overlaps, and sequential progress across all phases.

- This Gantt chart illustrates the project timeline from **September 2025 to October 2026**, highlighting key phases such as literature review, hardware setup, data collection, machine learning model development, system integration, testing, and documentation. It also includes extended phases for **system maintenance** and **iterative improvements**, ensuring the project evolves beyond initial deployment. Each task is visually mapped to its respective months, offering a clear view of dependencies and progress flow

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11.APPENDICES

Appendix A - Sensor Calibration and Validation Activities

This appendix summarizes the calibration procedures used to ensure accurate sensor measurements before data collection. Proper calibration improves the reliability of machine learning training datasets.

Category	Item Description	Present Study	Reference Study
Sensor Calibration	Accurately calibrate pH sensor before measurements	0.62	0.63
	Calibrate TDS sensor to ensure consistent readings	0.70	0.76
	Validate turbidity sensor with standard samples	0.63	0.64
	Verify conductivity sensor accuracy under different temperatures	0.54	0.73

Table 2 Sensor Calibration Procedures and Performance Comparison Between the Present Study and Reference Study

Appendix B - Water Quality Assessment Metrics

This appendix shows how water parameters are analyzed to determine the quality and classification of water samples.

Category	Item Description	Present Study	Reference Study
Water Quality Assessment	Measure pH and classify water type (distilled, tap, alkaline)	0.61	0.61
	Measure TDS and determine water suitability for consumption	0.48	0.50
	Measure turbidity and check compliance with safety standards	0.59	0.64
	Analyze temperature impact on water quality parameters	0.49	0.63

Table 3 Comparative Analysis of Water Quality Parameters Between the Present Study and Reference Study

Appendix C - Machine Learning Model Development and Evaluation

This appendix summarizes the machine learning approaches used for training and evaluating models for water classification.

Category	Item Description	Present Study	Reference Study
Data Analysis & ML Modeling	Train SVM model for water type classification	0.60	0.69
	Train KNN model for anomaly detection in water quality	0.65	0.70
	Train ANN for predicting TDS levels based on sensor inputs	0.68	0.74
	Evaluate model performance using cross-validation	0.71	0.75

Table 4 Performance Comparison of Machine Learning Models for Water Quality Analysis Between the Present Study and Reference Study

Appendix D - System Automation and IoT Communication

This appendix outlines the automation mechanisms and IoT communication features implemented in the system.

Category	Item Description	Present Study	Reference Study
System Planning & Automation	Develop automated water purification schedule based on sensor readings	0.63	0.74
	Implement alert system for unsafe water parameters	0.35	0.49
	Optimize IoT communication between sensors and cloud server	0.70	0.75

Table 5 Comparison of System Planning and Automation Performance in the Present Study and Reference Study

Appendix E - Troubleshooting and System Optimization

This appendix presents activities related to debugging system issues, improving model performance, and maintaining reliable sensor readings.

Category	Item Description	Present Study	Reference Study
Problem Solving & Troubleshooting	Detect sensor malfunctions and recalibrate	0.63	0.61
	Handle inconsistencies in water quality measurements	0.66	0.75
	Adjust purification process for varying water types	0.54	0.49
	Identify patterns in ML model misclassification	0.64	0.73
	Propose improvements for AI-based water quality prediction	0.61	0.57

Table 6 Evaluation of Problem Solving and Troubleshooting Effectiveness in Water Quality Monitoring Between the Present Study and Reference Study

Appendix F - Performance Comparison of Water Quality Monitoring and Machine Learning System Activities

Category	*Item Description*	*Present Study*	*Reference Study*
Sensor Calibration	*Accurately calibrate pH sensor before measurements*	*0.62*	*0.63*
	Calibrate TDS sensor to ensure consistent readings	*0.70*	*0.76*
Water Quality Assessment	*Validate turbidity sensor with standard samples*	*0.63*	*0.64*
	Verify conductivity sensor accuracy under different temperatures	*0.54*	*0.73*
	Measure pH and classify water type (distilled, tap, alkaline)	*0.61*	*0.61*
	Measure TDS and determine water suitability for consumption	*0.48*	*0.50*
Data Analysis & ML Modeling	*Measure turbidity and check compliance with safety standards*	*0.59*	*0.64*
	Analyze temperature impact on water quality parameters	*0.49*	*0.63*
	Train SVM model for water type classification	*0.60*	*0.69*
	Train KNN model for anomaly detection in water quality	*0.65*	*0.70*
System Planning & Automation	*Train ANN for predicting TDS levels based on sensor inputs*	*0.68*	*0.74*
	Evaluate model performance using cross-validation	*0.71*	*0.75*
	Develop automated water purification schedule based on sensor readings	*0.63*	*0.74*
Problem Solving & Troubleshooting	*Implement alert system for unsafe water parameters*	*0.35*	*0.49*
	Optimize IoT communication between sensors and cloud server	*0.70*	*0.75*
	Detect sensor malfunctions and recalibrate	*0.63*	*0.61*
	Handle inconsistencies in water quality measurements	*0.66*	*0.75*
	Adjust purification process for varying water types	*0.54*	*0.49*
	Identify patterns in ML model misclassification	*0.64*	*0.73*
	Propose improvements for AI-based water quality prediction	*0.61*	*0.57*

Table 7 Performance Comparison of Sensor Calibration, Water Quality Assessment, Machine Learning Modeling, System Automation, and Troubleshooting Activities Between the Present Study and Reference Study